

第四章 有心力 散射问题

§4.1 两体问题

采用78坐标：相对 (ρ, φ) 质心 (x_0, y_0, z_0)

* 两质点约束在平面上，减少了一个自由度

$$\text{势能 } V(\rho) = \int_{\rho}^{\infty} F(\rho) d\rho$$

$$\text{动能 } T = T_0 + T'$$

$$T_0 = \frac{1}{2} m (\dot{x}_0^2 + \dot{y}_0^2 + \dot{z}_0^2)$$

$$T' = \frac{1}{2} m_1 \dot{\vec{r}}_1'^2 + \frac{1}{2} m_2 \dot{\vec{r}}_2'^2$$

$$\vec{r}_1' = \frac{m_2}{m_1+m_2} \vec{r}, \quad \vec{r}_2' = -\frac{m_1}{m_1+m_2} \vec{r}$$

$$\Rightarrow T' = \frac{1}{2} m_1 \left(\frac{m_2}{m_1+m_2} \dot{\vec{r}} \right)^2 + \frac{1}{2} m_2 \left(-\frac{m_1}{m_1+m_2} \dot{\vec{r}} \right)^2$$

$$= \frac{1}{2} \frac{m_1 m_2}{m_1+m_2} \dot{\vec{r}}^2 = \frac{1}{2} m' \dot{\vec{r}}^2$$

$$\text{其中 } \vec{r} = \rho \hat{\rho}$$

$$T' = \frac{1}{2} m' [\dot{\rho}^2 + (\rho \dot{\varphi})^2]$$

$$L = T - V = \frac{1}{2} m_0 (\dot{x}_0^2 + \dot{y}_0^2 + \dot{z}_0^2) + \frac{1}{2} m' [\dot{\rho}^2 + (\rho \dot{\varphi})^2] - V(\rho)$$

① x_0, y_0, z_0, φ 是可逆坐标, 有 广义的动量守恒
即 φ 有对应恒定的速度不变.

$$P_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m' \rho^2 \dot{\varphi} = \text{const.} \Rightarrow \text{角动量守恒.}$$

$$\text{记 } h = \rho^2 \dot{\varphi}, \quad P_\varphi = m' h$$

② $\frac{\partial L}{\partial t} = 0$ 有 广义的能量守恒 $H = \sum p_i \dot{q}_i - L$

$$H = \frac{1}{2} m_0 (\dot{x}_0^2 + \dot{y}_0^2 + \dot{z}_0^2) + \frac{1}{2} m' (\dot{\rho}^2 + \rho^2 \dot{\varphi}^2) + V(\rho) = E$$

$$\text{同时: } \frac{1}{2} m' \left(\dot{\rho}^2 + \frac{h^2}{\rho^2} \right) + V(\rho) = E'$$

$$\text{记 } V_{\text{eff}}(\rho) = \frac{m' h^2}{2\rho^2} + V(\rho) \quad \text{则 } \frac{1}{2} m' \dot{\rho}^2 + V_{\text{eff}}(\rho) = E'$$

$$\text{解出 } \rho(t) : \quad \frac{d\rho}{\sqrt{\frac{2}{m'} [E' - V_{\text{eff}}(\rho)]}} = dt$$

$$\varphi(t) : \quad \varphi = \int \frac{h}{\rho^2} dt$$

③ 只需解 轨道方程 (使方程不显含 $t \Rightarrow$ 代换 $\frac{d}{dt}$)

$$\frac{d}{dt} = \frac{d}{d\varphi} \dot{\varphi} = \frac{d}{d\varphi} \left(\frac{h}{\rho^2} \right)$$

$$\text{代入: } \quad \frac{h d\rho}{\rho^2 \sqrt{\frac{2}{m'} [E' - V_{\text{eff}}(\rho)]}} = d\varphi$$

④ 轨道界限

$$\text{要求 } E' > V_{\text{eff}}(\rho)$$

$$\Rightarrow \begin{cases} \lim_{\rho \rightarrow \infty} V_{\text{eff}}(\rho) > E' & : \text{不允许 } \rho \rightarrow \infty, \text{ 有界运动} \\ < & : \text{允许 } \rho \rightarrow \infty, \text{ 无界运动} \end{cases}$$

④ Binet 公式

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{p}} - \frac{\partial L}{\partial p} = 0$$

$$\hookrightarrow m' \dot{p} - m' p \dot{\varphi}^2 + \frac{\partial V(\varphi)}{\partial p} = m' \dot{p} + \frac{\partial V(\varphi)}{\partial p} = 0$$

令 $u = \frac{1}{p}$ 接下来同法是求 $p(\varphi)$, $u(\varphi)$ 故要去掉 $\frac{d}{dt}$

$$\dot{p} = \dot{\varphi} \frac{dp}{d\varphi} = \dot{\varphi} \frac{d(\frac{1}{u})}{d\varphi} = -\frac{\dot{\varphi}}{u^2} \frac{du}{d\varphi} = -h \frac{du}{d\varphi}$$

$$\dot{p} = -h \dot{\varphi} \frac{d}{d\varphi} \frac{du}{d\varphi} = -h^2 u^2 \frac{d^2 u}{d\varphi^2}$$

$$\text{代入} \rightarrow -m' h^2 u^2 \frac{d^2 u}{d\varphi^2} - m' h^2 u^3 + \frac{\partial V}{\partial p} = 0$$

$$\text{即 比奈方程: } m' h^2 u^2 \left(\frac{d^2 u}{d\varphi^2} + u \right) = -F(\varphi)$$

若对相对数量守恒作变换

$$\frac{1}{2} m' [\dot{p}^2 + (p\dot{\varphi})^2] + V(\varphi) = E'$$

$$\rightarrow \frac{1}{2} m' \left[\left(-h \frac{du}{d\varphi} \right)^2 + h^2 u^2 \right] + V(\varphi)$$

$$= \frac{1}{2} m' h \left[\left(\frac{du}{d\varphi} \right)^2 + u^2 \right] + V(\varphi) = E'$$

考虑椭圆轨道 $p = \frac{P}{1 + \epsilon \cos \varphi}$ $u = \frac{1}{p} + \frac{\epsilon}{P} \cos \varphi$

$$F(\varphi) = -m' h \left(\frac{1}{p} + \frac{\epsilon}{P} \cos \varphi \right)^3 \left[\left(-\frac{\epsilon}{P} \cos \varphi \right) + \frac{1}{p} + \frac{\epsilon}{P} \cos \varphi \right]$$

$$= -m' h \frac{u^3}{P} = -\frac{m' h}{P p^3}$$

对左并反比力

§ 4.2 平方反比引力

1. Kepler 三定律 表明什么

$$\text{Kepler 1}^{st} : r = \frac{P}{1 + \varepsilon \cos \varphi}$$

$$r_{min} = \frac{P}{1 + \varepsilon}, \quad r_{max} = \frac{P}{1 - \varepsilon}, \quad a = \frac{P}{1 - \varepsilon^2}, \quad b = \frac{P \varepsilon}{1 - \varepsilon^2}, \quad d = \frac{P}{1 - \varepsilon^2}$$

$$\text{Kepler 2}^{nd} : \frac{1}{2} r^2 \dot{\varphi} = \frac{h}{2} = \text{const.} = C$$

$$\text{Kepler 3}^{rd} : \text{周期 } T = \frac{2\pi a b}{\frac{h}{2}} = \frac{2\pi}{h} P^2 (1 - \varepsilon^2)^{3/2}$$

$$\frac{C^2}{a^3} = \frac{4\pi^2 P}{h^2}$$

$$\hookrightarrow F = - \frac{m'}{P^2} h \cdot 4\pi^2 \frac{C^3}{C^2} \quad \text{书上是这样!} \quad \hat{=} - \frac{m}{P^2} 4\pi^2 h C \quad \text{表明平方反比力}$$

2. 平方反比力 表明什么

$$F = - \frac{mk}{\rho^2} \quad V(\rho) = \int_0^\infty F(\rho) d\rho = - \frac{mk^2}{\rho}$$

$$V_{eff} = V(\rho) + \frac{1}{2} m' \frac{h^2}{\rho^2} = - \frac{mk^2}{\rho} + \frac{m'h^2}{2\rho^2}$$

$$\text{径向上 } \frac{1}{2} m' \dot{\rho}^2 + V_{eff} = E'$$

$$\frac{d\rho}{\frac{2}{m'}(E' + \frac{mk}{\rho} - \frac{m'h^2}{2\rho^2})} = d\rho = d\varphi \cdot \frac{1}{\dot{\varphi}} = d\varphi \frac{\rho^2}{h}$$

$$\rightarrow \frac{h d\rho}{\rho \left(\frac{2}{m'} (\rho^2 E' + mk\rho - \frac{1}{2} m'h^2) \right)} = d\varphi$$

$$\text{而已: } \rho^2 E' + mk\rho - \frac{1}{2} m'h^2 = \left(\sqrt{E'} \rho + \frac{mk}{2\sqrt{E'}} \right)^2 - \frac{mk^2}{4E'} - \frac{1}{2} m'h^2$$

$$\text{令 } t = \sqrt{E'} \rho + \frac{mk}{2\sqrt{E'}} \rightarrow \int \frac{1}{\sqrt{E'}} \left(t - \frac{mk}{2\sqrt{E'}} \right) \sqrt{t^2 - \dots}$$

$$x(x^2+ax+b)$$

$$t = x + \frac{a}{2}$$

$$\int \sqrt{t - \frac{a}{2} + t^2 + c^2} \quad t = C \tan \varphi$$

$$C \tan \varphi = \frac{a}{2}$$

$$\hookrightarrow \arcsin \left[\frac{2m'k^2}{m'} \rho - 2h^2 \right] = \varphi + C + \frac{3}{2}\pi$$

$$\rho = \frac{2h^2}{\frac{2m'}{w}k^2 + \sqrt{\frac{4m'^2k^4 + 4E'h^2m'}{m'^2}}} = \frac{m'h^2}{mk^2 + \sqrt{m^2k^4 + 2E'h^2m' \cos(\varphi+C)}}$$

$$= \frac{\frac{m'h^2}{mk^2}}{1 + \sqrt{1 + \frac{2E'h^2m'}{m^2k^4} \cos(\varphi+C)}}$$

$$\text{令 } \rho = \frac{m'h^2}{mk^2} \quad \varepsilon = \frac{1}{1 + \frac{2E'h^2m'}{m^2k^4} \cos(\varphi+C)} \Rightarrow \rho = \frac{m'h^2}{mk^2} \varepsilon$$

$$\text{轨道方程: } a = \frac{1}{2} \left(\frac{P}{1-\varepsilon} + \frac{P}{1+\varepsilon} \right) = \frac{P}{1-\varepsilon^2} = \frac{\frac{m'h^2}{mk^2}}{\frac{m^2k^4}{2E'h^2m'}} = \frac{mk^2}{2|E'|}$$

$$b = a \sqrt{1-\varepsilon^2} = \frac{P}{1-\varepsilon^2} \quad \hookrightarrow |E'| = \frac{mk^2}{2a}$$

$$T = \frac{\pi ab}{\frac{1}{2}P^2} = \frac{2\pi ab}{h} = \frac{2\pi P^2}{h} (1-\varepsilon^2)^{-\frac{3}{2}}$$

$$\frac{T^2}{a^3} = \frac{4\pi^2}{h} P = \frac{4\pi^2}{h} \frac{m'}{m} \Rightarrow \text{平方反比力下运动满足开普勒三定律}$$

开普勒方程:

$$\frac{1}{2} m \dot{\rho}^2 = E' - V_{\text{eff}}(\rho)$$

$$\Rightarrow dt = \frac{d\rho}{\left| \frac{2E'}{m'} (\rho^2 + 2k^2\rho - h^2) \right|} = \frac{\rho d\rho}{\left| \frac{2E'}{m'} (\rho^2 + 2k^2\rho - h^2) \right|}$$

$$V_{\text{eff}} = \frac{1}{2} \frac{m' h^2}{\rho^2} - \frac{k^2 m'}{\rho}$$

$$k^2 = Gm$$

$$= \frac{\rho d\rho}{\left| \frac{2E'}{m'} (\rho^2 + \frac{k^2 m'}{E'} \rho - \frac{m' h^2}{2E'}) \right|^{1/2}}$$

$$|E'| = \frac{k^2 m'}{2a}$$

$$\Rightarrow a = \frac{k^2 m'}{2|E'|} = \rho$$

$$\textcircled{1} E' < 0$$

$$\left| \frac{m'}{2|E'|} \right| \rho d\rho$$

$$\int \frac{m'^2 k^4}{4|E'|^2} - \frac{m' h^2}{2|E'|} - \left(\rho - \frac{k^2 m'}{2|E'|} \right)^2 \textcircled{-\cos\zeta} = \left(\frac{m'^2 k^4}{4|E'|^2} - \frac{m' h^2}{2|E'|} \right)^{1/2}$$

$$\frac{m'^2 k^4}{4|E'|^2} - \frac{m' h^2}{2|E'|} = \left(\frac{k^2 m'}{2|E'|} \right)^2 \left(1 - \frac{2|E'| h^2}{k^4 m'} \right)$$

$$\Rightarrow a = \frac{k^2 m'}{2|E'|}, \quad \varepsilon = \left| 1 - \frac{2|E'| h^2}{m' k^4} \right|$$

$$\rho = a(1 - \varepsilon \cos\zeta) \quad d\rho = a \varepsilon \sin\zeta \, d\zeta$$

$$\int \left| \frac{m'}{2|E'|} \right| \frac{a^2 \varepsilon (1 - \varepsilon \cos\zeta) \sin\zeta \, d\zeta}{a \varepsilon \sin\zeta} = \int \left| \frac{m'}{2|E'|} \right| a (1 - \varepsilon \cos\zeta) \, d\zeta$$

$$\Rightarrow \text{开普勒方程: } t = \frac{a^{3/2}}{k} (\zeta - \sin\zeta) + t_0$$

$$\textcircled{2} E' = 0$$

$$dt = \frac{\rho d\rho}{|2k^2\rho - h^2|} \quad \zeta = |2k^2\rho - h^2| \quad \rho = \frac{1}{2k^2} (\zeta^2 + h^2)$$

$$= \frac{1}{\zeta} \frac{1}{2k^2} (\zeta^2 + h^2) \frac{\zeta}{k^2} d\zeta = \frac{1}{2k^4} (\zeta^2 d\zeta + h^2 d\zeta)$$

$$t = \frac{1}{6k^4} (\zeta^3 + 3h^2 \zeta) + t_0$$

③ $E' > 0$

$$dt = \frac{\rho d\rho}{\sqrt{\frac{2E'}{m'} \rho^2 + 2k^2 \rho - \hbar^2}} = \frac{\sqrt{\frac{m'}{2E'}}}{\sqrt{\frac{m'}{2E'}}} \left(\rho^2 + \frac{k^2 m'}{E'} \rho - \frac{m' \hbar^2}{2E'} \right)^{1/2}$$

$$= \frac{\sqrt{\frac{m'}{2E'}}}{\sqrt{\frac{m'}{2E'}}} \left[\left(\rho + \frac{k^2 m'}{2E'} \right)^2 + \frac{k^4 m'^2}{4E'^2} - \frac{m' \hbar^2}{2E'} \right]^{1/2}$$

$$a = \frac{k^2 m'}{2E'}, \quad \varepsilon = \left[1 - \frac{2E' \hbar^2}{k^4 m'} \right], \quad \rho + a = a \varepsilon \operatorname{ch} \zeta$$

$$t = \int \frac{\sqrt{\frac{m'}{2E'}}}{\sqrt{\frac{m'}{2E'}}} \frac{a \varepsilon \operatorname{sh} \zeta \{ \varepsilon \operatorname{ch} \zeta - 1 \}}{a \varepsilon \operatorname{sh} \zeta} d\zeta$$

$$= \frac{a^{3/2}}{k} \{ \varepsilon \operatorname{sh} \zeta - \zeta \} + t_0$$

3. 椭圆轨道的射量

4. 轨道的稳定性

$$\sim \frac{\partial^2 V_{eff}}{\partial \rho^2} \Big|_{\rho=\rho_0} > 0$$

(1) 圆轨道

① 有效势

$$m\ddot{\rho} = -\frac{\partial V_{eff}(\rho)}{\partial \rho} \quad \text{其中 } V_{eff}(\rho) = V(\rho) + \frac{mh^2}{2\rho^2}$$

$$\frac{\partial V_{eff}}{\partial \rho} = \frac{\partial V}{\partial \rho} + \frac{\partial}{\partial \rho} \left(\frac{mh^2}{2\rho^2} \right) = \frac{\partial V}{\partial \rho} - \frac{mh^2}{\rho^3}$$

$$\left. \frac{\partial V_{eff}}{\partial \rho} \right|_{\rho=\rho_0} = 0 \quad \Rightarrow \quad \left. \frac{\partial V}{\partial \rho} \right|_{\rho=\rho_0} = \frac{mh^2}{\rho_0^3}$$

$$\frac{\partial^2 V_{eff}}{\partial \rho^2} = \frac{\partial^2 V}{\partial \rho^2} + \frac{3mh^2}{\rho_0^5} \quad \text{其中 } \rho = \rho_0$$

$$\frac{\partial^2 V_{eff}}{\partial \rho^2} = -\left. \frac{\partial F}{\partial \rho} \right|_{\rho=\rho_0} + \frac{3}{\rho_0} \left. \frac{\partial V}{\partial \rho} \right|_{\rho=\rho_0} = -F'(\rho_0) - \frac{3}{\rho_0} F(\rho_0)$$

$$\text{小振动周期: } T = 2\pi \sqrt{\frac{m}{V_{eff}''(\rho_0)}}$$

② 线性化

$$\text{径向运动 } m(\ddot{\rho} - \rho\dot{\theta}^2) = F(\rho)$$

$$\text{角动量守恒 } \rho^2 \dot{\theta} = h$$

$$\ddot{\rho} + \frac{h^2}{\rho^3} = \frac{F(\rho)}{m} \quad \text{假设微扰: } \rho = \rho_0 + \delta$$

$$\hookrightarrow \ddot{\rho}_0 + \delta - \frac{h^2}{(\rho_0 + \delta)^3} = \ddot{\rho}_0 + \frac{h^2}{\rho_0^3} + \left(\ddot{\delta} + \frac{3h^2}{\rho_0^4} \delta \right) = \frac{F(\rho)}{m}$$

$$\rightarrow \ddot{\delta} = -\frac{3h^2}{\rho_0^4} \delta \quad \omega_\delta = \sqrt{\frac{3h^2}{\rho_0^4}}$$

例: 证明 $V = -\frac{nmk}{\rho^n}$ 对应的势函数 $n < 2$ 的稳定

$$F = -\frac{\partial V}{\partial \rho} = -\frac{nmk}{\rho^{n+1}}$$

$$\frac{\partial^2 V}{\partial \rho^2} = -\frac{n(n+1)mk}{\rho^{n+2}} + \frac{3}{\rho} \frac{nmk}{\rho^{n+1}} = \frac{nmk}{\rho^{n+2}} (2-n) > 0 \Rightarrow n < 2$$

例: 桌上小球, 分析

① 选取 ρ, φ 为广义坐标

$$\text{绳长} = l = \rho - z \rightarrow \dot{\rho} = \dot{z}$$

$$L = \frac{1}{2} m \dot{z}^2 + \frac{1}{2} m (\dot{\rho}^2 + \rho^2 \dot{\varphi}^2) - mgz$$

$$= m(\dot{\rho}^2 + \frac{1}{2} \rho^2 \dot{\varphi}^2) - mg\rho + mgl$$

φ 是可遗坐标 ($\frac{\partial L}{\partial \varphi} = 0$)

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = m \rho^2 \dot{\varphi} = \text{const.} = m h \quad \text{其中 } h = \rho^2 \dot{\varphi}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\rho}} - \frac{\partial L}{\partial \rho} = 2m\dot{\rho} - m\rho\dot{\varphi}^2 + mg = 0$$

$$\Rightarrow \ddot{\rho} = \frac{1}{2} \rho \dot{\varphi}^2 - \frac{1}{2} g = \frac{1}{2} \rho \left(\frac{h^2}{\rho^4} - g \right) = \frac{1}{2} \left(\frac{h^2}{\rho^3} - g \right)$$

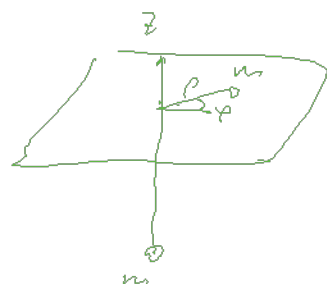
$$\ddot{\rho}|_{\rho_0} = 0 \Rightarrow \rho_0 = \sqrt[3]{\frac{h^2}{g}}$$

$$\text{令 } \rho = \rho_0 + \varepsilon \quad \ddot{\varepsilon} = \frac{1}{2} \left[\frac{h^2}{(\rho_0 + \varepsilon)^3} - g \right] \approx \frac{1}{2} \left[\frac{h^2}{\rho_0^3} \left(1 - \frac{3\varepsilon}{\rho_0} \right) - g \right]$$

$$= -\frac{3h^2}{2\rho_0^4} \varepsilon = -\frac{3}{2} g \sqrt[3]{\frac{h^2}{g}} \varepsilon = -\frac{3}{2} (gh)^{\frac{2}{3}} \varepsilon$$

$$\omega = \sqrt{\frac{3}{2} (gh)^{\frac{2}{3}}} = \frac{\sqrt{6}}{2} (gh)^{\frac{1}{3}}$$

② $E = T + V = m\dot{\rho}^2 + \frac{1}{2} \frac{m h^2}{\rho^2} + mg(\rho - l) = m\dot{\rho}^2 + V_{\text{eff}}(\rho)$



$$V_{eff}'(p_0) = 0 \Rightarrow p_0 = \sqrt[3]{\frac{\hbar^2}{g}}$$

$$\text{or: } \sqrt{\frac{V_{eff}'(p_0)}{m}} = \frac{\Gamma_0}{2} (g\hbar)^{\frac{1}{3}}$$

(2) 任意轨道

$$\text{体系方程: } m\hbar^2 u^2 \left(\frac{d^2 u}{d\varphi^2} + u \right) = -F(u)$$

假设原轨道为 $u_0(\varphi)$, 微扰后 $u(\varphi) = u_0(\varphi) + \delta(\varphi)$

$$m\hbar^2 (u_0^2 + 2u_0\delta + \delta^2) \left[\frac{d^2 u_0}{d\varphi^2} + \frac{d^2 \delta}{d\varphi^2} + u_0(\varphi) + \delta(\varphi) \right] = -F(u)$$

略去二阶小, 且 u_0 满足体系方程

$$m\hbar^2 \left\{ 2u_0\delta \cdot \left(\frac{d^2 u_0}{d\varphi^2} + u_0 \right) + u_0^2 \left[\frac{d^2 \delta}{d\varphi^2} + \delta(\varphi) \right] \right\} = 0$$

$$\left(\frac{d^2 u_0}{d\varphi^2} + u_0 \right) = -\frac{1}{m\hbar^2 u_0^2} F(u_0)$$

$$-\frac{2\delta}{u_0} F(u_0) + m\hbar^2 u_0^2 \left(\frac{d^2 \delta}{d\varphi^2} + \delta \right) = 0$$

$$m\hbar^2 u_0^2 \frac{d^2 \delta}{d\varphi^2} = - \left[m\hbar^2 u_0^2 - \frac{2F(u_0)}{u_0} \right] \delta$$

$$\text{小扰动频率} \frac{\omega}{\hbar} = \sqrt{\frac{m\hbar^2 u_0^2 - \frac{2F(u_0)}{u_0}}{m\hbar^2 u_0^2}}$$

§ 4.4 散射问题

1. 平方反比力下的运动

$$F(\rho) = \frac{k^2}{\rho^2} \quad V(\rho) = \int_{\rho}^{\infty} F(\rho) d\rho = \frac{k^2}{\rho}$$

$$\frac{1}{2} m' \dot{\rho}^2 + V(\rho) + \frac{m' h^2}{2 \rho^2} = E'$$

$$\frac{d\rho}{\sqrt{\frac{2}{m'}(E' - \frac{m' h^2}{2 \rho^2} - \frac{k^2}{\rho})}} = d\varphi \cdot \frac{1}{\dot{\varphi}} = \frac{\rho^2 d\varphi}{h} \rightarrow \frac{h d\rho}{\rho \sqrt{\frac{2E'}{m'} \rho^2 - h^2 - \frac{2k^2 \rho}{m'}}} = d\varphi$$

积分

$$\hookrightarrow \rho = \frac{1}{1 + \frac{1}{1 + \frac{2E' m' h^2}{k^4}} \cos(\varphi + \epsilon)} \quad \text{为双曲线}$$

$$\epsilon = \int \frac{1}{1 + \frac{2E' m' h^2}{k^4}} > 0$$

$(E' > V + \gamma > 0)$

或用比条方程:

$$m h^2 u^2 (\frac{d^2 u}{d\varphi^2} + u) = -k^2 u^2$$

$$\hookrightarrow \frac{d^2 u}{d\varphi^2} + u = -\frac{k^2}{m h^2}$$

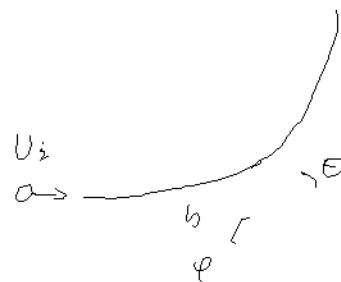
$$u = -\frac{k^2}{m h^2} + A \cos(\varphi - \varphi_0) \rightarrow \rho = \frac{1}{u} = \frac{\frac{m h^2}{k^2}}{A \frac{m h^2}{k^2} \cos(\varphi - \varphi_0) - 1} = \frac{P}{e \cos(\varphi - \varphi_0) - 1}$$

初始条件:

$$E = \frac{1}{2} m v_0^2, \quad h = v_0 b \quad \text{其中 } b \text{ 为瞄准距离}$$

取 $\theta_0 = 0 \rightarrow$ 核力心 - 远心点连线的极轴

$$\rho_{min} = \frac{P}{e-1}$$



$$E = \frac{1}{2} m (\dot{r}^2 + \frac{h^2}{r^2}) + \frac{k^2}{r}$$

$$\begin{aligned} \text{远心点} &= \frac{m h^2}{2 \rho_{min}^2} + \frac{k^2}{\rho_{min}} = \frac{m h^2 (e-1)^2}{2 P^2} + \frac{k^2}{P} (e-1) \\ &= \frac{k^4}{e^2} (e^2 - 1) \end{aligned} \quad P = \frac{m h^2}{k^2}$$

2mh

$$\text{解得 } e = \sqrt{1 + \frac{2Emh^2}{k^2}}$$

在原子核静止系中：

$$\rho \rightarrow \infty, \quad \cos \varphi \rightarrow 1 \quad \rightarrow \quad \begin{cases} \varphi_1 = \arccos \frac{1}{e} \\ \varphi_2 = -\arccos \frac{1}{e} \end{cases}$$

$$\text{散射角 } \theta = \pi - |\varphi_1| - |\varphi_2| = \pi - 2\arccos \frac{1}{e} = 2\left[\frac{\pi}{2} - \arccos \frac{1}{e}\right] = 2\arcsin \frac{1}{e}$$

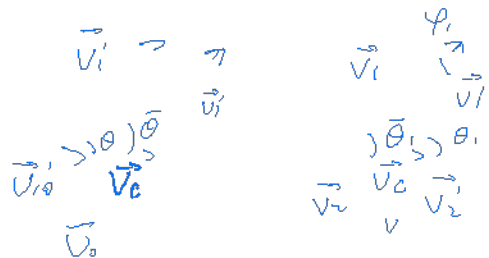
$$\sin \frac{\theta}{2} = \frac{1}{e} \quad e = \sqrt{1 + \frac{2Emh^2}{k^2}} \quad E = \frac{1}{2}mv_0^2, \quad h = v_0b$$

$$\hookrightarrow \cot \frac{\theta}{2} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \sqrt{1 + \frac{2Emh^2}{E^2}} \left[1 - \frac{1}{1 + \frac{2Emh^2}{E^2}} \right] = \frac{\sqrt{2Emh^2}}{k^2} = \frac{2Zh}{k^2}$$

换到实验室系中：粒子1以 v_0 入射

$$\vec{V}_C = \frac{m_1}{m_1+m_2} \vec{V}_0, \quad \vec{V}_0' = \vec{V}_0 - \vec{V}_C = \frac{m_2}{m_1+m_2} \vec{V}_0, \quad \vec{V}_{20}' = -\vec{V}_C = -\frac{m_1}{m_1+m_2} \vec{V}_0$$

$$\text{质心系 } C: \quad \vec{V}_{10}' \Rightarrow \vec{V}_1', \quad |\vec{V}_{10}'| = |\vec{V}_1'| \\ \vec{V}_{20}' \Rightarrow \vec{V}_2', \quad |\vec{V}_{20}'| = |\vec{V}_2'|$$



$$\text{实验室系 } L: \quad \vec{V}_0 \Rightarrow \vec{V}_0' = \vec{V}_C + \vec{V}_1' \\ 0 \Rightarrow \vec{V}_2' = \vec{V}_C + \vec{V}_2'$$

$$\tan \bar{\theta}_1 = \frac{V_1' \sin \theta_1}{V_C + V_1' \cos \theta_1} = \frac{m_2 \sin \theta_1}{m_1 + m_2 \cos \theta_1} = \frac{\eta \sin \theta_1}{1 + \eta \cos \theta_1}$$

$$\text{讨论 } \eta > 1, \text{ 即 } m_2 > m_1 \rightarrow \bar{\theta}_1 \in (0, \pi)$$

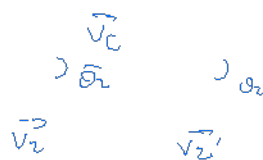
$$\eta < 1, \text{ 即 } m_2 < m_1 \rightarrow \bar{\theta}_1 \in (0, \arccos \eta)$$

$$\text{对2: } |\vec{V}_2'| = |\vec{V}_{20}'| = |\vec{V}_C|$$

$$\varphi_1 = \theta_1 - \bar{\theta}_1$$

$$\bar{\theta}_2 = \frac{\pi}{2} - \bar{\theta}_1$$

$$\hookrightarrow 2\bar{\theta}_2 = \bar{\theta}_2 = \pi - \bar{\theta}_1$$



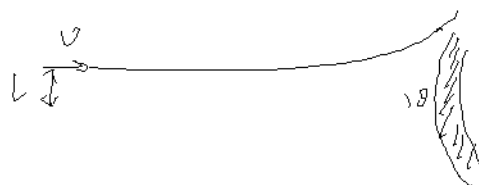
$\Rightarrow$ 实验显示中点高角: $\bar{\theta} = \bar{\theta}_1 + \bar{\theta}_2 = \frac{\pi}{2} + \frac{1}{2}(\bar{\theta}_1 - \varphi_1)$

$m=1$ 时: $|\vec{v}_1| = |\vec{v}_0| = |\vec{v}_c| \rightarrow \varphi_1 = \bar{\theta}_1$

$\bar{\theta} = \frac{\pi}{2}$

3. 散射截面

微分散射截面: $\sigma := \frac{dN}{n d\Omega}$



其中 dN 为单位时间散射到 $\Omega \sim \Omega + d\Omega$ 内粒子数

n 为入射粒子束单位时间、单位横截面入射粒子数

考察 $b \sim b + db$, $\theta \sim \theta + d\theta$ 的范围

$n \cdot 2\pi b db = n 2\pi \sin\theta d\theta \cdot \sigma$

$\Rightarrow \sigma(\theta) = - \frac{b db}{\sin\theta d\theta} \quad b \uparrow \theta \downarrow$

散射角满足: $\cot\left(\frac{\theta}{2}\right) = \frac{2E' b}{k^2}$

$\frac{db}{d\theta} = \frac{k^2}{2E'} \left(-\frac{1}{\tan^2\left(\frac{\theta}{2}\right)} \right) \cdot \frac{1}{2} = - \frac{k^2}{4E' \sin^2\left(\frac{\theta}{2}\right)}$

$\sigma(\theta) = - \frac{b}{\sin\theta} \frac{db}{d\theta} = \frac{k^2}{4E' \sin\theta \sin^2\left(\frac{\theta}{2}\right)} \cdot \frac{k^2 \cot\left(\frac{\theta}{2}\right)}{2E'} = \frac{k^4 \cot\left(\frac{\theta}{2}\right)}{8E'^2 \sin\theta \sin^2\left(\frac{\theta}{2}\right)}$

$= \frac{k^4}{16E'^2 \sin^4\left(\frac{\theta}{2}\right)}$

散射到某立体角内的粒子数与坐标无关

$dN(\theta) = n \sigma(\theta) d\Omega(\theta) \quad \bar{\theta} = \bar{\theta}(\theta)$

$= dN(\bar{\theta}) = n \sigma(\bar{\theta}) d\Omega(\bar{\theta})$

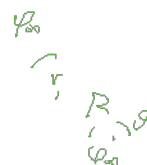
$\Rightarrow \frac{\sigma(\bar{\theta})}{d\Omega(\bar{\theta})} d\theta = \frac{\sin\theta}{\sin\bar{\theta}} d\theta$

总散射截面.

$$\Delta_T = \Delta(\theta) d\Omega(\theta) = \int_0^\pi 2\pi \sin\theta \Delta(\theta) d\theta$$

例：求一质量为 m 、半径为 r 的小弹子球与一个质量为 M ($m \ll M$)

半径为 R 的大弹子球散射角分布和总散射截面



$b \downarrow$

设瞄准距离为 l

$$\varphi_\infty = \arcsin\left(\frac{l}{R+r}\right)$$

$$\frac{d\theta}{dl} = -2 \frac{\frac{1}{R+r}}{1 - \left(\frac{l}{R+r}\right)^2}$$

$$\theta = \pi - 2\varphi_\infty = \pi - 2\arcsin\left(\frac{l}{R+r}\right)$$

$$\Delta(\theta) = \frac{dN}{n d\Omega} = - \frac{n \cdot 2\pi l dl}{n \cdot 2\pi \sin\theta d\theta} = -l \frac{dl}{\sin\theta d\theta}$$

$$= \int_0^l \sin\left[2\arcsin\left(\frac{l}{R+r}\right)\right] \cdot \frac{1}{2} \left[1 - \left(\frac{l}{R+r}\right)^2\right] (R+r)$$

$$= \int_0^l \left(\frac{l}{R+r} \left[1 - \left(\frac{l}{R+r}\right)^2\right]^{1/2}\right) \left[1 - \left(\frac{l}{R+r}\right)^2\right] (R+r)$$

$$= \frac{1}{4} (R+r)^2$$

$$\sigma_T = \int_0^\pi 2\pi \sin\theta \cdot \frac{1}{4} (R+r)^2 d\theta = \pi (R+r)^2$$